

Suavi Yildirim

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Summary

Mechanical Engineer and Master's candidate with 5+ years of experience in robotics and automation, in the final semester of the M.Sc. Autonomy Technologies at FAU.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg — Erlangen, DE
Master of Science, Autonomy Technologies — 04/2024 – Expected 03/2027

- Coursework completed; Master's thesis at Institute FAPS (submission end of October 2026)
- Thesis: Pneumatic levitation of wires using CFD simulation and machine-learning regression

Ozyegin University — Istanbul, TR
Bachelor of Science, Mechanical Engineering — 09/2017 – 07/2022

- 100% Merit Scholarship, Honor Student
- Graduated 1st in Mechanical Engineering (GPA 3.14/4.00, 1.8 German grade)

Experience

Institute for Factory Automation and Production Systems (FAPS) — 07/2024 – Present
Graduate Research Assistant (HiWi) — Nürnberg, DE

- Coordinated development and validation of an automated pneumatic wire transport system
- Integrated Siemens S7-1500 PLCs, HMI panels, and sensors in TIA Portal; tuned control parameters

Cobro Robot Technologies — 09/2023 – 05/2024
Concept Design Engineer – PreSales Robotics Automation — Istanbul, TR

- Translated customer requirements into robotic automation concepts for industrial B2B applications
- Developed a flexible feeder system for vision-guided handling of randomly oriented parts

Arche Robotics — 07/2020 – 08/2023
Mechanical Design Engineer – Autonomous Robotics — Istanbul, TR

- Technical team lead for autonomous mobile robots, from design to commissioning at customer sites
- Delivered two fully operational AMR systems in MediaMarkt stores

Projects & Awards

Siemens Healthineers Robotics & AI Hackathon | *Teach the Robot (VLA)* — 10/2026

- Robotics & simulation engineer (team of 6): simulation-based robot-learning loop for a bimanual Franka FR3 handover in Isaac Sim and Genesis; **12× faster** demo generation

Siemens x Sony Immersive Design Challenge | *BatteryTwin-XR* — 12/2024 – 04/2025

- Led Team NextCycle (XR digital twin for EV batteries); **1st place** of 900 participants

OzU Rover Team | *Ozyegin University* — 09/2017 – 04/2020

- Team lead (2018–2020); 13th of 56 teams at ERC 2019; two conference publications

Volunteering & Extracurricular Activities

- **Panelist and speaker** – sharing my experience in engineering and competitions at events and panels
- **Siemens Campus Connect Student Ambassador** (since 05/2025) – promoting Siemens software and events at FAU
- **Board Member, FAU Space Gene** – organizing events and activities for students
- **Event support at Institute FAPS** – organization and photography at the Besuchertage and the Digital | Power | Net opening

Skills

Technical: Siemens NX, CATIA, SolidWorks, CFD, Siemens S7-1500 / TIA Portal, ROS, NVIDIA Isaac Sim, Python, C++, MATLAB

Languages: English (professional working proficiency), German (B1)

Neumarkt i.d.OPf., October 25, 2026 — Suavi Yildirim

SUAVI YILDIRIM

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B.Sc. MECHANICAL ENGINEERING, OZYEGIN UNIVERSITY

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PORTFOLIO

Selected engineering projects in robotics, automation and product design — from mechanical design to simulation and robot learning.

PROJECT	ORGANISATION / PERIOD	HIGHLIGHT	PAGE
ROBOTICS & AI HACKATHON – TEACH THE ROBOT	Siemens Healthineers 10/2026	**12× faster demo generation**	3
IMMERSIVE DESIGN CHALLENGE	Siemens × Sony 12/2024 – 04/2025	**1st place of 900 participants**	3
PNEUMATIC WIRE TRANSPORTATION	FAU – Institute FAPS since 07/2024	**Stable wire transport at 20 m/s**	4
GRIPPER DESIGN FOR FOOD PACKAGING	Cobro Robot Technologies 2023 – 2024	**+20% throughput, −90% cost**	4
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TORSIONAL MEASUREMENT DEVICE	2E Design	**Anti-roll bar stiffness tuning**	8
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SUSPENSION CONTROL ARM OF A RALLY CAR	2E Design	**Stronger, more durable control arm**	8
SURGICAL ROBOT FOR AUTONOMOUS SUTURING	OzU Robotics Lab	**TÜBİTAK 1001 research project**	9
KUKA KR20-3 LIGHT PAINTING	Ozyegin University – RobArt 2021 – 2022	**Robot light painting**	9
KUKA KR20-3 PEN PLOTTER	Ozyegin University – RobArt 2021 – 2022	**Robot pen-plotter art**	9
THE HOPE	OzU Rover Team 2019 – 2020	**Top 36 of 102 teams (URC 2020)**	10
THEIA	OzU Rover Team 2018 – 2019	**13th of 56 teams (ERC 2019)**	10

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ROBOTICS & AI HACKATHON – TEACH THE ROBOT

SIEMENS Healthineers

Siemens Healthineers · 10/2026 · Robotics & Simulation Engineer, team of 6

Isaac Sim | Genesis | LeRobot | VLA | Quest 3







What?

- 3-day **Siemens Healthineers Robotics & AI Hackathon 2026** in Erlangen: build the **learning loop** for a **vision-language-action (VLA)** robot companion in the cathlab.
- Task: bimanual box handover with a **Franka FR3 Duo** in simulation.

How?

- **Robotics & Simulation Engineer** in a 6-person team.
- Built a **scripted IK expert** in **NVIDIA Isaac Sim 6.0** that generates demos with moved box poses (16/18 successful).
- Fine-tuned $\pi_{0.5}$ **(LoRA)** and **ACT**; rebuilt the task in **Genesis** for parallel data generation.

Results

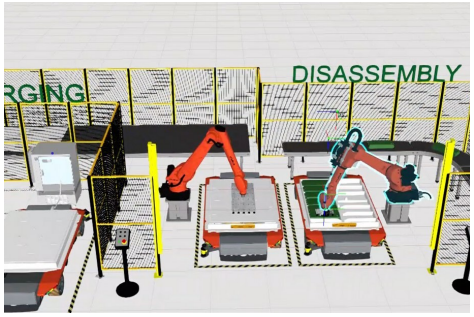
- Showed that the **data, not the model**, was the bottleneck: the 94 provided demos contain one single motion.
- **12× faster** demo generation: from 6 min to about 30 s per demo.
- Prototyped **VR teleoperation with Meta Quest 3** for human demos.

IMMERSIVE DESIGN CHALLENGE

SIEMENS | SONY

Siemens × Sony · 12/2024 – 04/2025 · Team lead

XR | Digital Twin | NX









What?

- The Immersive Design Challenge is a **global engineering competition** organized by Siemens Digital Industries Software in collaboration with Sony.
- Encourages **sustainable design** by bringing immersive technologies such as **XR and digital twins** into real engineering problems.
- Scale: ∼**900 students**, **230+ universities**, **38 countries**.

How?

- **Led** the team **NextCycle**.
- **Developed** BatteryTwin-XR: a Siemens-integrated **digital twin** for a sustainable **EV battery lifecycle**.
- **Designed** an XR framework with the Sony XR head-mounted display to support **battery monitoring, reuse and recycling** decisions.
- Focused on **circular economy**: better recycling efficiency, less battery waste.

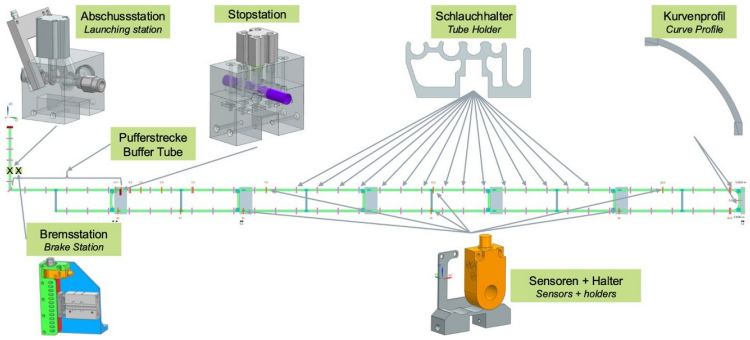
Results

- **Won 1st place** among the international finalist teams.
- Recognized by Siemens and Sony for an **innovative, sustainability-driven** XR solution.
- Awarded a **Sony XR headset** and a 1-year **NX Immersive Designer** license for academic and research use.
- Invited to **Siemens Realize LIVE**; helped expand immersive design capabilities at FAU.

PNEUMATIC WIRE TRANSPORTATION

FAU – Institute FAPS · since 07/2024 · Graduate research assistant

Siemens NX | 3D printing | Pneumatics





What?

- As part of **Project SIMON** at the **Institute FAPS**, I contributed to an **automated pneumatic "shooting" system** that replaces manual **wire handling** and removes assembly bottlenecks for flexible wire components.

How?

- Used **Siemens NX** for CAD modeling of the **experimental setup**, including tube holder, stop and brake stations.
- Used **FDM and SLA 3D printing** for parts with different **structural** and high-tolerance **aerodynamic** requirements.

Results

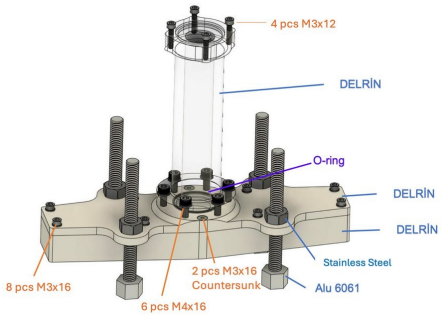
- **Achieved stable transport** at **20 m/s** while keeping a defined state of order for flexible leads.
- **Quantified** empirical pressure-loss coefficients for industrial-scale pneumatic system modeling.

GRIPPER DESIGN FOR FOOD PACKAGING

Cobro Robot Technologies · 2023 – 2024 · Concept design engineer

ABB Delta | Vacuum gripping







What?

- **Developed** robot grippers for **ABB Delta robots**, tailored to the operational and hygiene requirements of a high-speed **food production company**.

How?

- Designed grippers for high-speed **pick-and-place** operations.
- **Optimised gripper mass** to meet the robot payload limits.
- **Used O-ring sealing** for reliable, repeatable suction performance.

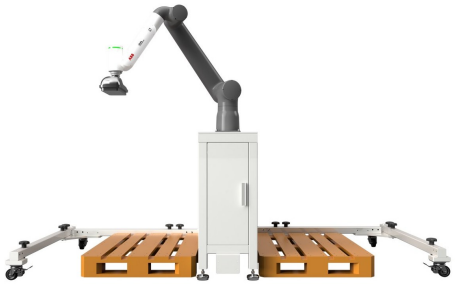
Results

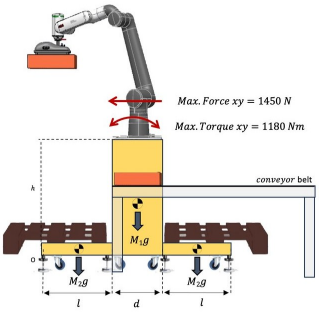
- **Integrated** into the production line: **20% higher** packaging throughput.
- **90% cost reduction** by replacing standard components with custom suction cups.

COBOT PALLETIZER

Cobro Robot Technologies · 2023 – 2024 · Concept design engineer

ABB CRB 15000 | Cobot | Structural analysis







What?

- **Developed** a **mobile palletizing robot** with collaborative robots that moves easily between **production lines** for more flexibility and efficiency.

How?

- Performed **static and dynamic analyses** to design a robust base fixture.
- Selected the **ABB CRB 15000** cobot with a **12 kg payload**.

Results

- The system was **successful** and helped the company win further projects, as it stands out from stationary palletizers on the market.

FLEXIBLE FEEDER SYSTEM

Cobro Robot Technologies · 2023 – 2024 · Concept design engineer

Machine vision | Bin picking



What?

- A **flexible feeding system** for manufacturers that need robotic automation to **separate mixed products**.





Results

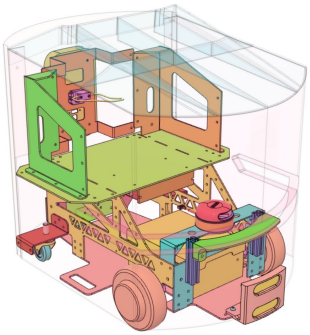
- A **cost-effective** automated feeder that sorts mixed parts, ready to be extended with **image processing and AI**.

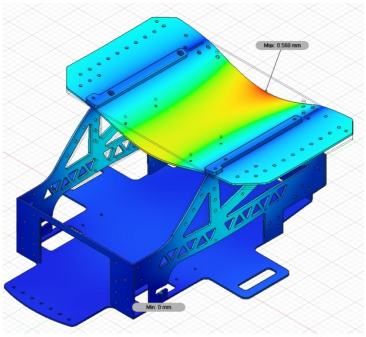
AT1 MOBILE PLATFORM

ARCHE ROBOTICS

Arche Robotics · 2020 – 2023 · Mechanical design engineer

SolidWorks | Fusion 360 | FEA







What?

- **AT1** is an indoor **autonomous mobile robot** platform (AMR).
- A versatile platform for **multiple applications**.
- Designed the chassis for a **50 kg** payload.

How?

- Used **SolidWorks** and **Fusion 360** for the mechanical design.
- Ran an **FEA study** of the forces and moments on the chassis.
- 3 mm **6061 aluminium** sheet-metal parts joined by TIG welding.

Results

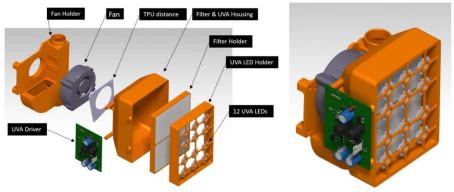
- **Launched** AT1, now used from **healthcare** to **retail** and **hospitality**.
- Basis for ongoing development of new hardware and software modules.

PURA

ARCHE ROBOTICS

Arche Robotics · 2020 – 2023 · Mechanical design engineer

Photocatalysis | Air sensing







What?

- **PURA** is based on the AT1 platform and **sanitizes** surfaces and floors and **purifies the air**.
- Designed a sanitation chamber with a **photocatalyst module** and **HOCl disinfection** liquid.
- Senses **air quality**: particles, VOC, humidity, temperature.

How?

- Photocatalyst module with **16 ceramic foams** coated with TiO_2 and **40 UVA LEDs**.
- A **piezo fog** generator makes a **dry fog** of disinfectant, blown by a DC fan at **150 m^3/h**.
- No consumables such as HEPA filters, cartridges or mercury UVC bulbs.

Results

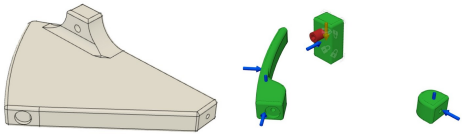
- **Deployed** an autonomous disinfection robot that works **24/7**.
- Cleans as much air per hour as **300 people** breathe.
- Works **without harmful chemicals**.
- Built strategic relationships with Chinese manufacturers.

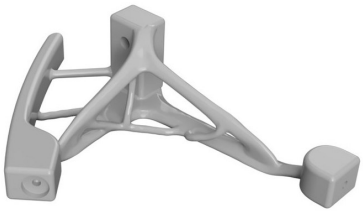
GENERATIVE DESIGN STUDY

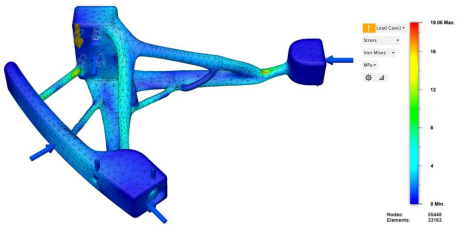
ARCHE ROBOTICS

Arche Robotics · 2020 – 2023 · Mechanical design engineer

Fusion 360 | Generative design







What?

- Used **generative design** (AI-driven algorithms) to create **lightweight parts** that meet performance and design requirements.

How?

- Used the **Fusion 360** generative design toolbox.
- Defined the **loads** on the part; compared **von Mises stress** results to ensure structural integrity.

Results

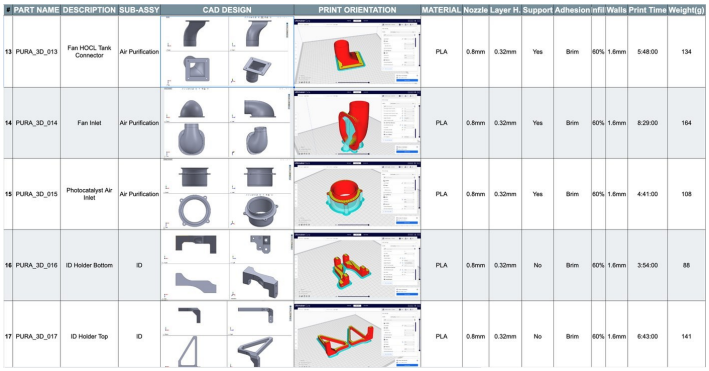
- **30% less weight** with all design constraints and performance criteria met.

ADDITIVE MANUFACTURING PART MANAGEMENT

ARCHE ROBOTICS

Arche Robotics · 2020 – 2023 · Mechanical design engineer

FDM | Part tracking



	PART NAME	DESCRIPTION	SUB-ASSY	CAD DESIGN	PRINT ORIENTATION	MATERIAL	Nozzle	Layer H.	Support	Adhesion	infil	Walls	Print Time	Weight(g)
13	PURA_3D_013	Fan HOCL Tank Connector	Air Purification			PLA	0.8mm	0.32mm	Yes	Brim	60%	1.6mm	5:48:00	134
14	PURA_3D_014	Fan Inlet	Air Purification			PLA	0.8mm	0.32mm	Yes	Brim	60%	1.6mm	8:29:00	164
15	PURA_3D_015	Photocatalyst Air Inlet	Air Purification			PLA	0.8mm	0.32mm	Yes	Brim	60%	1.6mm	4:41:00	108
16	PURA_3D_016	ID Holder Bottom	ID			PLA	0.8mm	0.32mm	No	Brim	60%	1.6mm	3:54:00	88
17	PURA_3D_017	ID Holder Top	ID			PLA	0.8mm	0.32mm	No	Brim	60%	1.6mm	6:43:00	141

What?

- Created an **Excel-based tracking system** to document 3D-printed parts and keep an organized **inventory**.

Results

- **Standardized** part quality through identical slicing settings.
- **Efficient tracking** of series production, total print time and material use.

PROMOBOT

ARCHE ROBOTICS

Arche Robotics · 2020 – 2023 · Mechanical design engineer

AMR | Retail | Dashboards







What?

- Promobot helps businesses **engage customers** and build their brand in physical and digital stores.
- Features: **welcoming**, **customer guidance**, **marketing** and **AR applications**.

How?

- A **data-driven** companion that supports **digital transformation**.
- Monthly dashboards of sensor data such as **air quality** and **demographics**.

Results

- **Deployed** at **MediaMarkt** Metropol Istanbul on 2 Jan 2023.
- Interacted with **400 customers** per month on average.
- Drove **10+ km** autonomously per month without mechanical issues.

SUAVI YILDIRIM

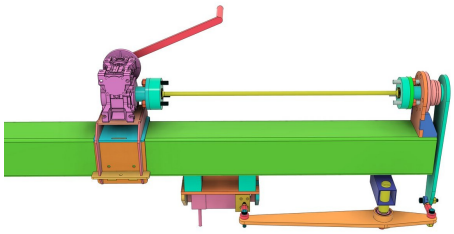
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TORSIONAL MEASUREMENT DEVICE

2E Design · Mechanical design

Test rig · Load cell · Motorsport







What?

- **Developed** a torsional measurement device to analyse the mechanical characteristics of **anti-roll bars**.

How?

- Hub and bearing setup to measure anti-roll bar **twist**.
- Load transferred through lever arms to a **1-ton load cell**.

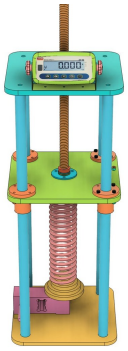
Results

- Enabled measuring and **tuning** anti-roll bar stiffness for racing vehicles.

SPRING STIFFNESS MEASUREMENT DEVICE

2E Design · Mechanical design

Test rig · Motorsport



What?

- **Developed** a device to measure the **spring stiffness** of racing-car **suspension springs**.

Results

- Accurate **spring coefficients**, used to tune the suspension for **optimal track performance**.

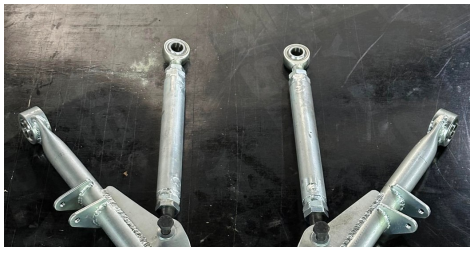


SUSPENSION CONTROL ARM OF A RALLY CAR

2E Design · Mechanical design

Redesign · Manufacturing · Motorsport







What?

- R&D to improve the **suspension control arm** of a rallycross car for performance and durability under racing conditions.

How?

- **Redesigned** the arm after the original broke during a race.
- **Manufactured** a stiffer, more robust arm and fitted it to the car.

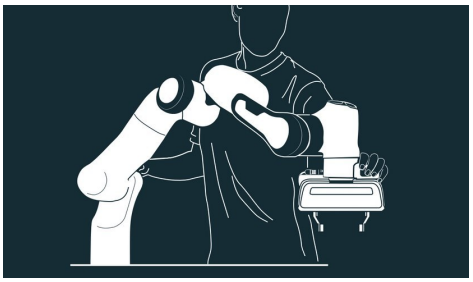
Results

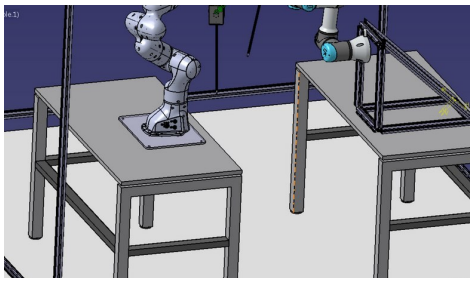
- A **stronger, more durable** arm that improved performance and reliability on the track.

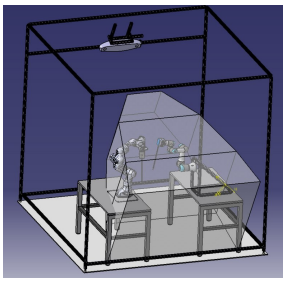
SURGICAL ROBOT FOR AUTONOMOUS SUTURING

OzU Robotics Lab · Mechanical design

CATIA V5 | UR3 | Franka Panda







What?

- A **surgical robot** with two cobots, **UR3** and **Franka Panda**. I designed the **end-effector flanges** for the Franka Panda arms and the structure for the **3D vision sensors**.

How?

- Modeled robots and sensors in **CATIA V5**.
- Adapted the **Da Vinci EndoWrist** as end-effector.
- **Built** tables from welded rigid profiles to reduce vibration.

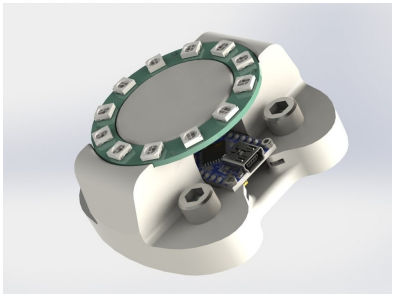
Results

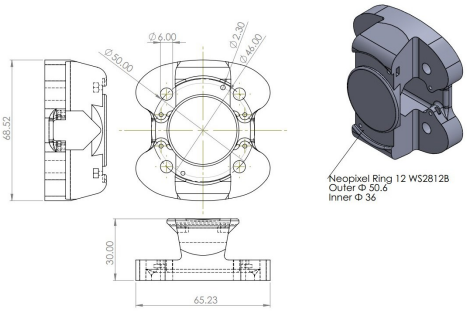
- Ongoing at the **OzU Robotics Laboratory**: “Development of Surgical Robot Assistants for Autonomous Suturing with Skill Transfer from Surgeon to Robot”, funded by **TÜBİTAK 1001**.

KUKA KR20-3 LIGHT PAINTING

Ozyegin University – RobArt · 2021 – 2022

KUKA | Rhino/Grasshopper





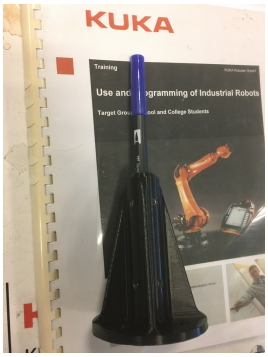
What?

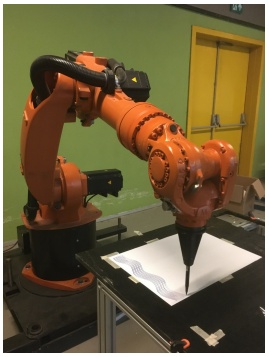
- In my senior year at Ozyegin University I took part in the **RobArt** workshops with industrial robot arms.
- **Light painting**: programmed a **KUKA KR20-3** with **Rhino/Grasshopper** and photographed the planned trajectory.
- **Designed and 3D printed** a flange holding an **LED ring**.

KUKA KR20-3 PEN PLOTTER

Ozyegin University – RobArt · 2021 – 2022

KUKA | 3D printing





What?

- A second RobArt workshop on **pen-plotter art**: the robot arm draws 2D images.
- **Designed and 3D printed** a flange holding a pen on the KUKA KR20-3.

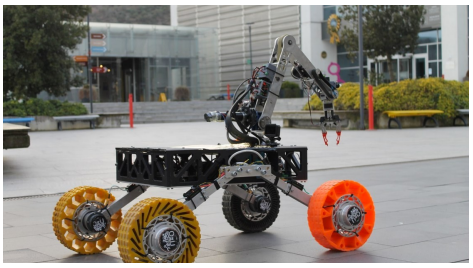
THE HOPE

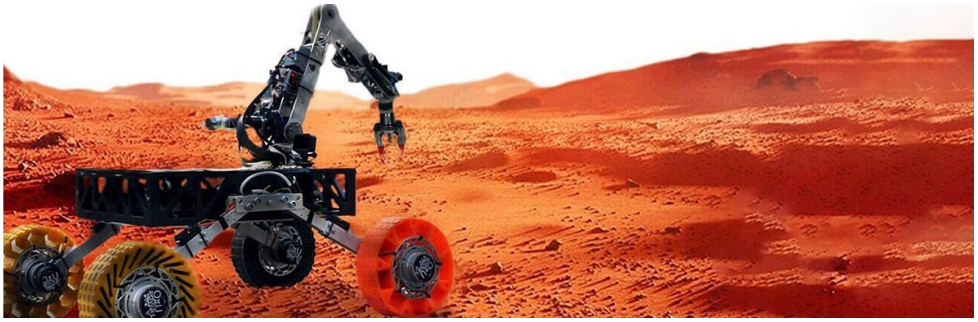
OzU Rover Team · 2019 – 2020 · Team lead

Mars rover URC









What?

- **The Hope** is a Mars rover prototype built by the OzU Rover Team for the **University Rover Challenge 2020** in the USA.

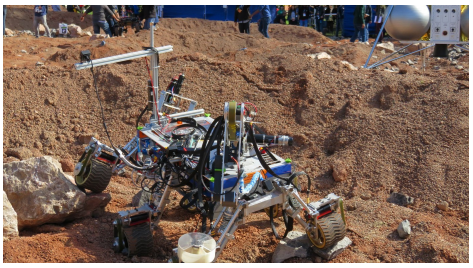
Results

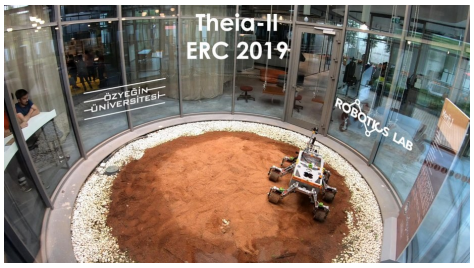
- **Qualified**, ranking in the top 36 of 102 teams.
- The event was cancelled due to COVID-19.

THEIA

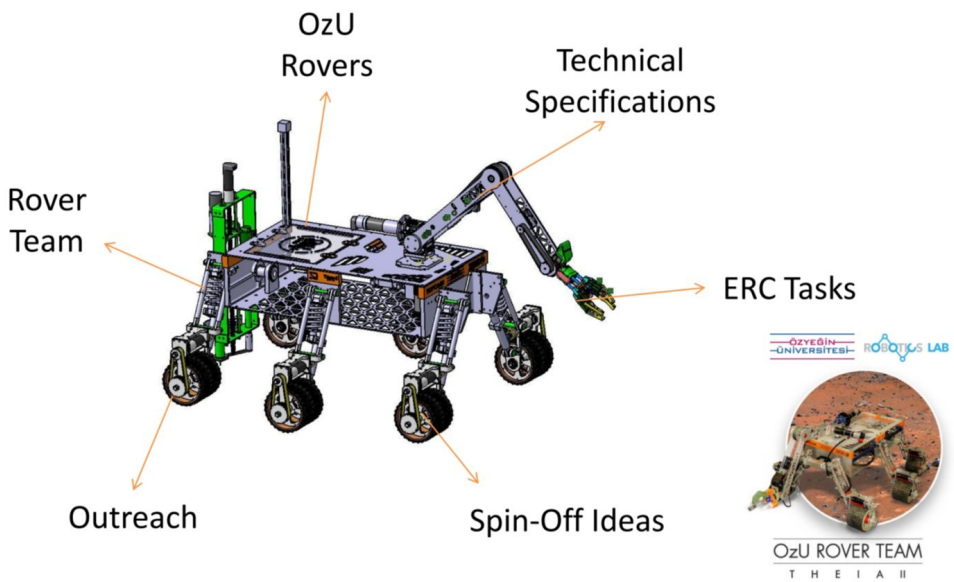
OzU Rover Team · 2018 – 2019 · Mechanical designer, then team lead

Mars rover ERC









What?

- Theia was built by the OzU Rover Team for the **European Rover Challenge**.
- Teams design and build a **Mars rover prototype** for **tele-operated** or **autonomous** tasks that assist astronauts.

Results

- **ERC 2018: 26th/65** as mechanical designer.
- **ERC 2019: 13th/56** as **team lead**, through leadership and design optimization.